

EACMR-IoT

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Chapter 1

Introduction

The rapidly growing Internet of Things (IoT) is transforming businesses and everyday life around the world. From smart homes to industrial-made appliances, IoT devices facilitate flawless communication and data-driven decision-making. As there are more and more devices being connected together, the importance of improved efficiency, enhanced security, and driven innovation continues to grow for IoTs [6]. The widespread adoption of these Internet of Things devices has reshaped how we engage with technology, resulting in a smarter and more linked society.

One of the most important components in enabling this communication and interaction, especially in resource-limited settings[13], is the **Routing Protocol for Low-Power and Lossy Networks**. This protocol is responsible for managing data flow in networks where devices have limited power, bandwidth, and processing capacity, making it a widely adopted protocol for IoT deployments [7]. The hierarchical structure, which is based on **Destination-Oriented Directed Acyclic Graph (DODAG)** formation and dynamic routing capabilities [5], means if there is a node that is unavailable due to energy depletion, it can automatically reroute through other available nodes without disrupting the traffic, making it an essential tool for optimising network performance in IoT applications such as healthcare, industrial automation, and smart cities.

However, the rapid growth of IoT networks has also exposed critical security weaknesses within the RPL protocol. IoT devices, often deployed in large, distributed, and low-power environments, are increasingly at risk of data breaches, unauthorised access, and other cyber threats [2]. The lack of built-in security features in RPL makes these networks particularly vulnerable to attacks aimed at compromising the confidentiality and authenticity of data.

This proposal paper proposes a methodology for defining an effective authentication and confidentiality mechanism in RPL-based IoT networks, where the critical Contiki-NG operating system is deployed in upcoming IoT devices from various sectors, such as smart

homes, health care, environmental monitoring, and, most critically and concerning for us, the military sector. By integrating lightweight cryptographic algorithms such as Elliptic Curve Cryptography (ECC), Simon, and Speck, the project aims to strengthen both the authentication and confidentiality of data transmitted within RPL-based IoT networks. These cryptographic mechanisms are designed to offer robust security while maintaining computational efficiency, which guarantees that they are suitable for the resource-constrained nature of IoT devices [10]. Through this approach, the project aims to mitigate the weaknesses within RPL, providing a more secure framework for IoT communication.

1.1 Problem Domain

This study will examine the critical security challenges faced by **constrained IoT devices** operating within **RPL-based networks**. The **Internet of Things (IoT)** has rapidly expanded across various sectors, embedding intelligence into everyday devices to enable communication and data exchange. Nevertheless, the resource limitations of IoT devices (processing power, memory and energy) are real critical barrier today when it comes to the adoption of traditional security mechanisms. These limitations create weaknesses in the areas of **authentication** and **confidentiality**, which are vital for secure operation of IoT networks, especially in sensitive environments like healthcare, smart cities, and industrial automation.

At the heart of many IoT networks is the **Routing Protocol for Low-Power and Lossy Networks (RPL)**, which was designed to facilitate efficient communication among devices in resource-constrained environments [10]. While RPL excels in managing data flow and optimising network resources, it lacks robust built-in security features. This creates several **security gaps**, particularly in the areas of **data integrity, authentication, and confidentiality**. As a result, RPL-based IoT networks are highly vulnerable to attacks such as actors can exploit the lack of secure authentication mechanisms to gain unauthorized access or disrupt network operations.

The primary gap in this domain is the lack of **lightweight cryptographic solutions** tailored to the specific needs of constrained IoT devices [5]. This leads to weakened security or, at worst case, a lack of security as traditional cryptographic algorithms are often too computationally expensive over such devices. Moreover, RPL networks' poor key management can be exploited to steal keys and prevent forward/backward secrecy which is important because devices join/leave frequently [5]. As such, this work addresses a critical gap in prior research by examining use of lightweight cryptographic mechanisms delivering strong authentication and privacy whilst being low-footprint to the constrained resources of IoT devices.

1.2 Research Problem Statement

The exact problem addressed in this research is the absence of efficient and lightweight security mechanisms for RPL-based IoT networks, specially when it comes to authentication and confidentiality. With the expansion of constrained IoT devices into areas like health care and industrial manufacturing, many security solutions are still too resource-intensive, leaving networks susceptible to routing attacks that can leave data interceptions and unauthorized access pathways open. However, the **RPL protocol** while great at dealing with low power communication does not have built in security measures to an effective shield against such threats [5]. This creates a significant gap in ensuring secure data transmission and the reliable authentication of devices, particularly as IoT networks scale and evolve.

This research aims at bridging this gap by designing a lightweight security mechanism that uses **Elliptic Curve Cryptography (ECC)** for authentication and the **Simon** and **Speck algorithms** for confidentiality. The study aims to develop this cryptographic solutions for **RPL** based networks, considering the limitations of IoT devices and providing a framework that will provide security without affecting network performance or efficiency. This approach will address the critical need for scalable, secure communication in resource-constrained[10] environments, providing a solution that balances security with computational and energy limitations. The significance of this research lies in its potential to enhance the **resilience of IoT networks** to external attacks, ensuring a safer and more reliable infrastructure for interconnected devices in various real-world applications.

1.3 Software Requirements Specification (if applicable)

Functional Requirements

1. Authentication Mechanism:

- The software must implement **Elliptic Curve Cryptography (ECC)** to authenticate devices within the network.

2. Confidentiality Mechanism:

- The software must use **Simon/Speck algorithms** to encrypt data transmitted between IoT devices and the network, ensuring data confidentiality.
- The system must generate and manage **dynamic encryption keys** that change as nodes join or leave the network to maintain **forward and backward secrecy**.

3. Key Management:

- The system must support **dynamic key revocation** and **key reassignment** to ensure the security of data and devices as the network topology changes.
- Secure storage and distribution of cryptographic keys must be implemented, ensuring minimal exposure to attacks during key exchange or revocation.

4. Quality of Service (QoS) :

- Routing mechanism should be efficient in terms of resources (memory, bandwidth, processing), scalable, and have low latencies and delays.

Non-Functional Requirements

1. Performance and Efficiency:

- The implementation must maintain **low computational overhead**, ensuring that security mechanisms do not overburden the limited processing capabilities of IoT devices.
- The solution should operate within the **power and memory constraints** of constrained IoT devices, ensuring that devices can run securely without significantly reducing battery life or network performance.

2. Compatibility:

- The software must be compatible with existing **IoT architectures** and protocols, ensuring smooth integration with existing RPL-based networks.

3. Reliability:

- The system must provide **high availability** and **fault tolerance**, ensuring that security services remain operational even if some nodes experience failure or leave the network.

4. Latency and Response Time:

- The system must introduce **minimal latency** to network communications, ensuring that the addition of security mechanisms does not hinder the real-time operation of IoT networks.
- The **response time for authentication** and encryption processes must be optimized to maintain seamless communication between IoT devices and the network.

Chapter 2

Literature Review

2.1 Related Research

2.1.1 Fuzzy Elliptic Curve Cryptography for Authentication in Internet of Things

2.1.1.1 Summary of the research item:

The paper titled "**Fuzzy Elliptic Curve Cryptography for Authentication in Internet of Things**" [1] addresses the crucial need for robust security and privacy measures in IoT systems. With the increasing integration of intelligent technology into our daily lives, the paper[1] emphasises the importance of protecting the data exchanged between end-users and sensing devices. The authors propose a secure and lightweight authentication protocol tailored for IoT systems, leveraging Elliptic Curve Cryptography (ECC) enhanced by fuzzy logic for random number generation. The protocol aims to bolster both authentication and encryption processes, thereby enhancing the overall security framework of IoT systems.

2.1.1.2 Critical analysis of the research item:

Strengths:

1. **Relevance and Importance of Topic:** The paper addresses the critical need for secure and lightweight authentication protocols in IoT systems, which is increasingly important due to the growing dependence on smart technology.
2. **Performance Comparison:** A key strength is the use of elliptic curve cryptography (ECC) combined with random numbers generated by fuzzy logic. The paper

shows that this approach outperforms existing ECC methods based on shift registers, providing empirical evidence of its effectiveness.

3. **Security Evaluation:** The paper enhances its analysis by testing the proposed scheme against attack algorithms like Pollard's p and Baby-step Giant-step, offering a deeper security evaluation that identifies potential vulnerabilities.

Weaknesses:

1. **Lack of Evaluation Against Advanced Threats:** The paper lacks specific details on how the proposed method is evaluated against advanced threats.
2. **Omission of Limitations and Challenges:** The paper does not discuss potential limitations or challenges in impact of the protocol on computational efficiency and power consumption

2.1.1.3 Relationship to the proposed research work

The research on "**Fuzzy Elliptic Curve Cryptography for Authentication in Internet of Things**"[1] is highly relevant to the proposed work, as both focus on securing resource-constrained IoT systems using lightweight cryptographic methods. Both studies share a common objective of enhancing the security of IoT systems, particularly for resource-constrained devices. The paper's use of **ECC with fuzzy logic**[1] for improving key generation aligns with our focus on **ECC for key management**, making it a relevant comparison. Furthermore, the emphasis on lightweight cryptographic protocols in both works highlights the importance of balancing security with efficiency.

2.1.2 A Lightweight Certificate-Based Authentication Scheme for 6LoWPAN-Based Internet of Things

2.1.2.1 Summary of the research item:

The paper, "**A Lightweight Certificate-Based Authentication Scheme for 6LoWPAN-Based Internet of Things**"[1], addresses the security challenges faced by low-power IoT devices. The authors propose a lightweight authentication scheme that effectively addresses the resource constraints of low-power IoT devices by utilising cryptographic hash functions and elliptic curve cryptography (ECC) for efficient authentication and key exchange. The authors implement a certificate-based system for device authentication, where certificates are issued by a gateway node (GWN). The proposed scheme aims to mitigate several well-known attacks, including replay, man-in-the-middle, and device impersonation. The AVISPA tool verifies the scheme's security, and BAN logic confirms

its correctness. The paper[11] also performs a comparative analysis to demonstrate that the scheme achieves a better trade-off between security and communication/computation overhead compared to other relevant schemes.

2.1.2.2 Critical analysis of the research item:

Strengths:

1. **Well-suited for the resource-constrained nature of IoT devices:** as it uses **ECC**, which offers strong security with lower key sizes, and lightweight **hash functions** to minimize computational overhead.
2. **Utilising tools such as AVISPA and BAN logic:** for formal verification and correctness proof, which strengthens the credibility of the proposed method.
3. **Mitigates several key attacks:** including replay and man-in-the-middle attacks, making it robust for deployment in open and insecure networks typical of IoT systems.

Weaknesses:

1. **Lack of real-world testing** data on actual devices, which could provide a more concrete understanding of its efficiency in various scenarios.
2. **Limited discussion on scalability**, which is a crucial factor for IoT networks with large numbers of devices.

2.1.2.3 Relationship to the proposed research work

The proposed research on **Efficient Authentication and Confidentiality Mechanism for RPL-Based IoT Networks (EACMR-IoT)** is closely related to the paper, as both works aim to secure **resource-constrained IoT environments** using **Elliptic Curve Cryptography (ECC)** for authentication. While the paper[11] focusses on certificate-based authentication for resource constrained devices, the proposed research builds on similar principles by using ECC for robust authentication and key management in RPL-based IoT networks. The certificate-based approach in the paper[11] can inform the **device registration** and key management system of the proposed work, where IoT devices generate their own public keys and securely register with an authentication server or border router. Furthermore, both solutions emphasise **lightweight cryptographic techniques**; however, the proposed research extends beyond ECC by integrating **SIMON and SPECK lightweight block ciphers** for **data confidentiality**, ensuring a comprehensive security framework

that addresses both **authentication** and **encryption**. Additionally, the paper's focus on mitigating attacks like replay and impersonation informs the proposed solution's features, such as **forward and backward privacy**, which aim to protect past and future sessions even if a current session key is compromised. Lastly, the **AVISPA** and **BAN logic** verification tools used in the paper could be adapted to validate the security and correctness of the protocol being developed in the proposed research.

2.1.3 An Efficient Lightweight Privacy-Preserving Mechanism for Industry 4.0 Based on Elliptic Curve Cryptography

2.1.3.1 Summary of the research item:

The paper titled "An Efficient Lightweight Privacy-Preserving Mechanism for Industry 4.0 Based on Elliptic Curve Cryptography" [12], proposes a secure authentication and data sharing mechanism tailored for Industry 4.0 environments. Given the resource limitations of IoT devices in industrial settings, the study focusses on developing a lightweight cryptographic solution that leverages Elliptic Curve Cryptography (ECC) for mutual authentication and data encryption. The paper[12] introduces a certificate-based system for validating device identities and securing communications, using ECC for its smaller key sizes and lower computational overhead. The scheme also counters a variety of cyberattacks such as replay, eavesdropping, and impersonation attacks. The paper[12] presents a performance evaluation, demonstrating reduced computational and communication costs when compared to existing methods. The AVISPA tool formally verifies the system's security, confirming its robustness against key security threats.

2.1.3.2 Critical analysis of the research item:

Strengths:

1. **Elliptic Curve Cryptography (ECC)**, with its lower computational demands and smaller key sizes, makes the proposed solution highly suitable for resource-constrained IoT devices
2. **Thorough security evaluation using the AVISPA tool**, which reinforces the credibility of the system's resilience to key attacks such as replay, impersonation, and eavesdropping.

Weaknesses:

1. **Lacks real-world deployment and testing in industrial IoT environments**, which could provide deeper insights into its performance under varying network conditions and at a larger scale.
2. **The scalability not explored in depth**: While the ECC-based approach is efficient for smaller networks, its performance in large-scale IoT deployments with thousands of devices should be further investigated.

2.1.3.3 Relationship to the proposed research work:

The proposed research on developing an **Efficient Authentication and Confidentiality Mechanism for RPL-Based IoT Networks (EACMR-IoT)** shares several similarities with the work presented in this paper. Both studies leverage **Elliptic Curve Cryptography (ECC)** for lightweight and efficient **authentication** in **resource-constrained environments**. While the paper focuses on **Industry 4.0** [12], the proposed research applies similar ECC-based cryptographic methods to secure **RPL-based IoT networks**. Furthermore, the **certificate-based authentication** mechanism presented in the paper could inform the key management approach in the proposed research, particularly in securely distributing and revoking public keys. Additionally, the security verification techniques used in the paper [12], such as **AVISPA**, can be adopted in the proposed work to formally evaluate the robustness of the proposed system against various attacks. The integration of **SIMON and SPECK** lightweight block ciphers in the proposed research also complements the lightweight design principles emphasized in the paper [12], allowing for a more comprehensive solution that ensures both **authentication** and **confidentiality** in IoT networks.

2.1.4 Text Encryption Using Pell Sequence and Elliptic Curves with Provable Security

2.1.4.1 Summary of the research item:

The paper titled "**Text Encryption Using Pell Sequence and Elliptic Curves with Provable Security**" [3] describes a new way to encrypt text messages that is meant to meet the growing need for safe communication in computer and network environments. The proposed scheme uses the Pell sequence and elliptic curves in a three-step process: diffusion of the plaintext, encoding of the diffused text using a combination of Pell sequences and binary sequences, and confusion through permutations over elliptic curves. The authors believe that the scheme provides strong resistance against a range of modern cryptographic attacks, including key sensitivity, statistical attacks, and known plaintext attacks,

which guarantees security. The paper [3] concludes with a detailed comparison to existing encryption methods, discussing the significance of the proposed method in terms of security against modern cryptanalysis techniques.

2.1.4.2 Critical analysis of the research item:

Strengths:

1. **Innovative Use of Pell Sequence and Elliptic Curves:** introduces a unique combination of Pell sequences and elliptic curves for text encryption.
2. **Provable Security:** The authors provide a comprehensive security analysis, this demonstrates that the method proposed by the author has provable security against multiple types of cryptanalytic attacks, including key spacing, ciphertext-only, and known-plaintext attacks.

Weaknesses:

1. **Complexity in Implementation:** The encryption scheme's highly complicated usage of Pell sequences and elliptic curves presents a considerable challenge in its implementation, especially in situations with limited resources.
2. **Performance Overhead:** The multi-step encryption process, including diffusion, encoding, and confusion, likely introduces significant computational overhead, which could impact the performance, especially in real-time or high-throughput systems.
3. **Scalability Issues:** The scheme's dependence on complex mathematical operations may not scale well with larger datasets or more extensive networks, potentially limiting its applicability in scenarios requiring high scalability.

2.1.4.3 Relationship to the proposed research work:

The research that is being proposed on **Efficient Authentication and Confidentiality Mechanism for RPL-Based IoT Networks (EACMR-IoT)** is similar in some ways to the way that "Text Encryption Using Pell Sequence and Elliptic Curves with Provable Security"[3] talks about encryption. Both approaches leverage **Elliptic Curve Cryptography (ECC)** to ensure data security, highlighting the importance of elliptic curves in modern cryptography for resource-constrained environments. While the paper[3] introduces an innovative use of **Pell sequences** alongside ECC, the proposed research focusses on the lightweight encryption provided by **SIMON and SPECK** ciphers, which are more suitable for IoT devices due to their lower computational overhead.

The multi-step encryption process in the Pell-ECC scheme[3] is an intriguing parallel to the layered security in the proposed research, which integrates robust ECC-based authentication with lightweight encryption. But the Pell-ECC method has problems with performance and scalability, mainly computational overhead and scalability limits. This is different from the EACMR-IoT method, which focusses on making IoT networks more resource efficient in the real world. The proposed research aims to mitigate these limitations by ensuring that the cryptographic methods chosen (SIMON, SPECK, and ECC) offer scalability and low energy consumption, which are critical for **large-scale IoT deployments**.

2.1.5 Cryptography Algorithms for Enhancing IoT Security

2.1.5.1 Summary of the research item:

The research article "Cryptography Algorithms for Enhancing IoT Security" [9] extensively discusses the role of lightweight cryptographic algorithms, particularly Simon and Speck, in improving IoT device security. Simon and Speck are block ciphers developed by the NSA specifically designed to address the security needs of resource-constrained environments, such as IoT, where devices have limited processing power, memory, and energy. The main focus of the study is on how these ciphers can effectively provide authentication and confidentiality, which are critical security requirements in IoT applications.

Simon and Speck [4] are highly optimised for their respective implementations: Simon for hardware and Speck for software. Simon uses a simple structure based on bitwise operations, rotations, and modular additions, making it efficient for devices like RFID tags, sensors, and embedded systems that rely on minimal computational power. While Speck employs simple operations, it is more versatile in software environments and offers better performance in terms of speed and flexibility. The study highlights that these ciphers maintain a balance between security and efficiency, making them ideal for IoT applications where lightweight yet robust encryption is essential to protect against unauthorised access and data breaches.

2.1.5.2 Critical analysis of the research item:

Strengths

1. **Efficiency for IoT Devices:** The simplicity of **Simon and Speck** ciphers makes them highly suitable for IoT environments, as they require minimal computational overhead, memory, and power consumption which are critical factors for resource-constrained devices.

2. **Hardware and Software Adaptability:** **Simon** excels in hardware due to its efficient use of bitwise operations, while **Speck** is optimised for software, providing versatility across different IoT applications, with customisable key and block sizes for varying levels of security and performance.
3. **Low Power Consumption:** **Simon's** design, focused on logical shifts and bitwise operations, ensures that devices with limited battery life, such as RFID tags and sensors, can operate securely without draining resources.

Weaknesses

1. **Scepticism Due to NSA Origins:** Despite no evidence of backdoors, the public perception of Simon and Speck ciphers being developed by the NSA introduces scepticism and mistrust in their use, which could impact their adoption in certain sectors.
2. **Scalability issues:** as the time moves on, its difficult to manage and update security protocols.
3. **Trade-offs:** The limited resources of LWC lead to security compromises.

2.1.5.3 Relationship to the proposed research work:

The proposed research on **Efficient Authentication and Confidentiality Mechanism for RPL-Based IoT Networks (EACMR-IoT)** aligns closely with the findings in this study. Both approaches recognise the value of **lightweight cryptographic solutions** like **Simon and Speck** for securing **resource-constrained IoT devices**. The proposed work can leverage Simon and Speck's strengths, such as their **low computational overhead** and **power efficiency**, to ensure confidentiality in IoT data transmissions, particularly in environments where power and processing resources are limited. However, the weaknesses identified in the study, such as the **lack of real-world benchmarking**, highlight the need for empirical testing in the proposed research to validate Simon and Speck's performance on actual IoT devices. In addition, adding authentication protocols to the privacy offered by Simon and Speck will fill in the gaps found in this study, making sure that the proposed IoT network has a full end-to-end security solution.

2.1.6 Lightweight cryptography in IoT networks: A survey

2.1.6.1 Summary of the research item:

The paper titled "**Lightweight Cryptography in IoT Networks: A Survey**[8]" by Muhammad Rana et al. provides a comprehensive review of the existing cryptographic solutions

developed for securing IoT systems, particularly focussing on lightweight cryptography. The research highlights the limitations of traditional cryptographic protocols such as AES and RSA when applied to IoT environments due to their high computational demands. The paper[8] categorises lightweight cryptography into symmetric (block and stream ciphers) and asymmetric algorithms like elliptic curve cryptography (ECC). It presents a comparative analysis of popular lightweight block ciphers like PRESENT, SIMON, and RECTANGLE, focussing on their efficiency in terms of key size, block size, gate area, and resistance to attacks. The study[8] emphasises that IoT devices, being resource-constrained, require cryptographic algorithms that are not only secure but also highly efficient in power consumption, latency, and memory usage. The research concludes by identifying the need for developing more efficient cryptographic solutions tailored for resource-limited IoT environments.

2.1.6.2 Critical analysis of the research item:

Strengths

1. **Comprehensive Review:** The paper provides a **thorough comparison** of state-of-the-art lightweight cryptographic algorithms, such as **PRESENT, SIMON, and RECTANGLE**, based on performance metrics like memory consumption, latency, and throughput, making it a valuable resource for researchers in **IoT security**.
2. **Algorithm Efficiency:** It highlights the **efficiency of lightweight block ciphers** like PRESENT and SIMON in **hardware implementations**, and RECTANGLE's **resilience** against cryptanalytic attacks, offering insights into the balance between security and performance in constrained environments.
3. **Balanced View of Trade-offs:** The paper effectively discusses the **trade-offs** between security, performance, and resource constraints, providing a balanced view of how these factors affect IoT systems.

Weaknesses

1. **Lack of Real-World Testing:** The effectiveness of these ciphers in practical IoT deployments is not well-validated, with a lack of empirical data from real-world applications.
2. **Implementation Challenges:** There is minimal discussion on the **challenges of integrating** these cryptographic algorithms into **real-world IoT systems**, especially in **hybrid systems** involving both hardware and software components.

3. **Key Management and Distribution:** While it emphasizes lightweight cryptography, the paper does not delve deeply into **key management and distribution**, which are essential for ensuring overall security in IoT networks.

2.1.6.3 Relationship to the proposed research work:

This paper[8] talks about lightweight cryptographic algorithms like **SIMON**, which is also part of the proposed solution. The proposed research on **Efficient Authentication and Confidentiality Mechanism for RPL-Based IoT Networks (EACMR-IoT)** fits with what this paper says. The look at performance metrics like memory usage and latency is helpful for making the proposed research's use of **SIMON and SPECK** to protect IoT devices with limited resources work better. There are some problems with the current research, like not enough real-world testing and problems with integrating it. These show that the proposed research can make progress by testing the cryptographic algorithms in real IoT systems, especially in RPL-based networks. Also, the paper doesn't talk much about **key management and distribution**, but the proposed research can fill in this gap by creating a strong **key management strategy** that protects IoT networks' security without slowing them down.

2.1.7 Lightweight Cryptography Algorithms for Resource-Constrained IoT Devices

2.1.7.1 Summary of the research item:

The paper "**Lightweight Cryptography Algorithms for Resource Constrained IoT Devices: A Review, Comparison and Research Opportunities**"[10] focusses on lightweight cryptography algorithms specifically designed for resource-constrained IoT devices. These algorithms are essential for IoT applications that involve low-power devices like RFID tags, sensors, and smart cards. The study[10] provides a comparison of over fifty lightweight cryptography algorithms based on performance metrics such as memory, gate area, power consumption, and resistance to attacks. Simon and Speck, designed by the NSA, are praised for their compact design and low computational cost, making them suitable for both hardware and software implementations. The paper[10] concludes by identifying the challenges of balancing security, performance, and cost in lightweight cryptography and calls for further research into new solutions that can better meet these requirements.

2.1.7.2 Critical analysis of the research item:

Strengths:

1. **Hardware and Software Efficiency:** It highlights the efficiency of Simon and Speck, distinguishing between their hardware and software implementations. Simon's minimal gate count for hardware and Speck's high throughput for software environments are well-addressed.
2. **Energy Efficiency:** The paper underlines Simon's suitability for energy-efficient environments, an essential consideration for IoT devices that prioritize low power consumption.
3. **Use of ARX Operations:** It explains the significance of ARX (Addition-Rotation-XOR) operations in Speck, which is an important cryptographic feature balancing speed and simplicity in software.

Weaknesses:

1. **Lack of Real-World Benchmarking:** The paper could benefit from more practical, real-world benchmarks or examples of the algorithms in action, particularly in IoT environments. The discussion remains theoretical without performance metrics.
2. **Focus on Efficiency Over Security:** The emphasis on the efficiency of Simon and Speck may lead readers to overlook the importance of their security trade-offs, which are essential for certain IoT applications where security is paramount.
3. **Insufficient Discussion on Alternatives:** The paper discusses AES, PRESENT, Simon, and Speck but leaves out other lightweight cryptographic algorithms that might offer different trade-offs in specific IoT environments, such as LEA or HIGHT.

2.1.7.3 Relationship to the proposed research work:

As research work aims to achieve Efficient Authentication and Confidentiality Mechanism for RPL-based IoT Networks. For that purpose we need lightweight cryptographic algorithms. The above paper[10] helps us analyze the algorithms and which one to choose to achieve more efficient results. As around 50 LWC algorithms were compared and evaluated based on many factors. This helps us finding the best ones for our research and work on it.

2.1.8 IoT Security: Implementation of Xtea, Simon/Speck Lightweight Block Ciphers

2.1.8.1 Summary of the research item:

The paper "**IoT Security: Implementation of Xtea, Simon/Speck Lightweight Block Ciphers**" [4] by Levent Ertaul and Arpit Chauhan examines the performance of Xtea, Simon, and Speck block ciphers on resource-constrained IoT devices. Given the limited memory, processing power, and energy of devices like microcontrollers used in IoT applications, the study highlights the need for lightweight encryption. The authors implemented and tested these ciphers on an ESP8266 microcontroller, focussing on execution time, power consumption, and memory usage. Simon and Speck, developed by the NSA, were designed for constrained environments, with Simon optimised for hardware and Speck for software applications, both excelling in low-resource settings.

2.1.8.2 Critical analysis of the research item:

1. **Practical Evaluation:** The study provides real-world insights by testing Simon, Speck, and Xtea ciphers on an ESP8266 microcontroller, offering valuable data for IoT applications where devices have limited memory, processing power, and energy.
2. **Flexibility and Scalability:** Simon and Speck are highlighted for their adaptability, supporting various block and key sizes, making them suitable for a wide range of IoT devices, from RFID tags to more complex embedded systems.
3. **Hardware and Software Suitability:** Simon's optimization for hardware and Speck's for software make these ciphers versatile, broadening their use cases within IoT environments.

Weaknesses:

1. **Performance Trade-offs:** Simon and Speck are less efficient than Xtea in terms of execution speed and memory usage. Their additional operations (rotations, AND, XOR) increase computational complexity, leading to slower encryption times and higher power consumption.
2. **Limited Security Discussion:** The study lacks an in-depth analysis of the ciphers' resistance to cryptographic attacks, which is crucial for evaluating their security robustness in IoT environments.

3. **Key Management and Protocol Integration:** The paper could benefit from further discussion on implementation challenges, particularly around key management, which is critical for maintaining security in IoT systems with minimal human intervention.

2.1.8.3 Relationship to Proposed Research:

The proposed research on the **Efficient Authentication and Confidentiality Mechanism for RPL-Based IoT Networks (EACMR-IoT)** shares a common focus with the study on lightweight cryptography for constrained IoT environments. Both works highlight the use of **Simon and Speck** for ensuring data confidentiality, emphasizing their efficiency in low-resource settings. However, the research[10] aims to address the **performance trade-offs** identified in the study by optimising their integration within the RPL protocol, balancing flexibility with better performance. Furthermore, the proposed work complements the study by incorporating **Elliptic Curve Cryptography (ECC)** for robust authentication, addressing key management challenges, and improving overall security, an area the study touches upon but doesn't explore in depth.

2.2 Analysis Summary of research items

Below is the Tabular representation with discussion of the critical analysis of the research items discussed in the previous section of this chapter.

Table 2.1: Literature Review Comparison

Research Paper	Year	Domain	Evaluation Metrics	Benefits	Limitations
Fuzzy Elliptic Curve Cryptography for Authentication in Internet of Things [1]	14 Jan 2022	IoT Security and Cryptography.	Authentication accuracy, Randomness quality, Encryption strength, Performance	1. Performance comparison with existing methods 2. Tested against attack algorithms	1. Lack of Evaluation Against Advanced Threats 2. Impact on resources
A Lightweight Certificate-Based Authentication Scheme for 6LoWPAN-Based Internet of Things [11]	15 Mar 2023	IoT Security and Cryptography	Authentication strength, Computation/communication overhead, Correctness verification, Scalability	1. Well-suited for the resource-constrained 2. Utilising tools such as AVISPA and BAN logic 3. Mitigates several key attacks	1. Lack of real-world testing data 2. Limited discussion on scalability,
Continued on next page					

Table 2.1 – continued from previous page

Research Paper	Year	Domain	Evaluation Metrics	Benefits	Limitations
An Efficient Lightweight Privacy-Preserving Mechanism for Industry 4.0 Based on Elliptic Curve Cryptography[12]	31 Dec 2021	IoT Security and Cryptography	Computational efficiency, Communication overhead, Scalability, Real-world applicability	1. ECC lower Computational demand (small key size) 2. Thorough security evaluation	1. Lacks real-world deployment and testing 2. The scalability not explored in depth
Text Encryption Using Pell Sequence and Elliptic Curves with Provable Security [3]	14 Jan 2022	Text Encryption, Cryptography, Elliptic Curve Cryptography	Encryption strength, Computational complexity, Scalability, Performance overhead	1. Introduces a unique combination of Pell sequences and elliptic curves 2. provable security against multiple types of cryptanalytic attacks	1. Complexity in Implementaiton specially with limited resources. 2. Performance Overhead 3. Scalability Issues
Continued on next page					

Table 2.1 – continued from previous page

Research Paper	Year	Domain	Evaluation Metrics	Benefits	Limitations
Cryptography Algorithms for Enhancing IoT Security[9]	23 March 2023	IoT Security, Lightweight Cryptography	Computational efficiency, Power consumption, Security strength, Hardware and software adaptability , Scalability	1. Efficiency for IoT Devices 2. Low power consumption 3. Hardware and Software Adaptability	1. Scalability issues 2. Trade-offs
Lightweight cryptography in IoT Networks: A survey [8]	27 Nov 2021	IoT Security, Lightweight Cryptography	Computational efficiency, Power consumption, Security robustness, Scalability, Key management and distribution	1. Use of performance matrices 2. Highlights Balance between security and performance 3. Algorithm Efficiency	1. Lack of Real-World Testing 2. Implementation Challenges 3. Does not delve into Key Management and Distribution
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Table 2.1 – continued from previous page

Research Paper	Year	Domain	Evaluation Metrics	Benefits	Limitations
Lightweight Cryptography Algorithms for Resource-Constrained IoT Devices[10]	19 January 2021	Lightweight Cryptography, Security	Memory usage, Computational efficiency, Power consumption, Security robustness, Real-world performance	1. Use of ARX Operations 2. Prioritises low power consumption 3. Highlights the efficiency of Simon and Speck	1. Lack of Real-World Benchmarking 2. Insufficient Discussion on Alternatives 3. Focus on Efficiency Over Security
IoT Security: Implementation of Xtea, Simon/Speck Lightweight Block Ciphers[4]	Im- 09 April 2024	IoT Security, Lightweight Cryptography	Power Consumption, Memory Usage, Flexibility and Scalability	1. Provides real world insights 2. Flexibility and Scalability 3. Hardware and Software Suitability	1. Performance Trade-offs 2. Lacks an in-depth analysis of the ciphers' resistance to cryptographic attacks 3. Key Management and Protocol Integration

Chapter 3

Proposed Approach

The solution will be implemented in the RPL Protocol which is a Routing Protocol for Low Power and Lossy networks below is the working 3.1. The term Lossy networks mean a high packet drop rate. In IoT devices have low power and Lossy networks, so RPL protocol is widely used. The RPL is mainly used with IPv6 but can also implemented in IPv4. The RPL is a distance vector routing protocol, it is based on Destination Oriented Directed Acyclic Graphs (DODAG). Above is the working of the protocol of how it works, where DODAG is created and nodes are joining in through messages.

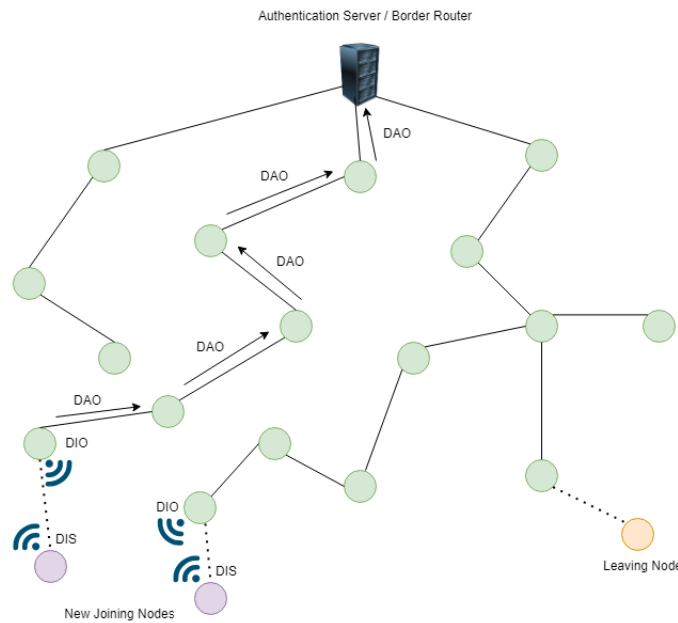


Figure 3.1: Working of RPL Protocol

To address the identified security vulnerabilities in RPL-based IoT networks, the proposed solution involves the development of an **Efficient Authentication and Confidentiality Mechanism for RPL-Based IoT Networks (EACMR-IoT)**. The solution will

leverage **Elliptic Curve Cryptography (ECC)** for robust authentication and the **Simon and Speck lightweight block ciphers** for ensuring data confidentiality. The proposed system will be implemented in phases as seen in the components diagram below [3.2](#), including:

1. **Enhancing RPL Protocol Security:** The proposed approach revolves around the development of the Secure RPL (SRPL) protocol, specially tailored to address the security challenges encountered by constrained IoT devices utilizing the RPL protocol. This will be done in RPL code in the Contiki-NG operating system. This will involve different cryptographic mechanisms discussed in below points.
2. **Robust Registration and Authentication:** An authentication server or border router will ensure that the mitigation of external attacks by integration of **Elliptic Curve Cryptography (ECC)** is used for robust authentication, key management, and revocation. ECC is selected for its efficiency in resource constrained environments due to its smaller key size. The Secure distribution of global parameters will also be done from this server among IoT devices through DIO message (RPL Protocol). Each Iot device will be responsible to generate its own public key and share it with its **Unique Device Identification** to server with DAO message. This will result in device registration as it will store correspond ID with its public key. Once this is done then, server will send a nonce to the registered node, when this nonce is received by the registered nonce it will use digital signature (encrypt the nonce with its private key) and send it back to the server to authenticate itself, as it can only be decrypted by the public key of registered node. Now the node is authenticated.
3. **Forward and Backward Privacy:** The design will also include feature which will ensure both forward and backward privacy. Forward privacy will safeguard against the compromise of current session keys from enabling unauthorised access to past sessions. Meanwhile, backward privacy will be implemented to prevent disclosure of future session keys even if a current session key is compromised.
4. **Confidentiality:** Our solution will integrate the lightweight cryptographic algorithm such as SIMON and SPECK, ensuring the protection of sensitive data within the constrained IoT network. The SIMON and SPECK will be used to generate symmetric key which is passed on to each authenticated node for maintaining the confidentiality of data transmissions. The symmetric key is encrypted through the public key of each node and sent to them so that each node, that can decrypt it using their own private key. This further ensures high security.
5. **Session Creation:** Secure communication sessions will be established using dynamically managed encryption keys. This process involves generating and distributing encryption keys securely to the nodes, making sure that each session maintains confidentiality and is protected against unauthorized access.

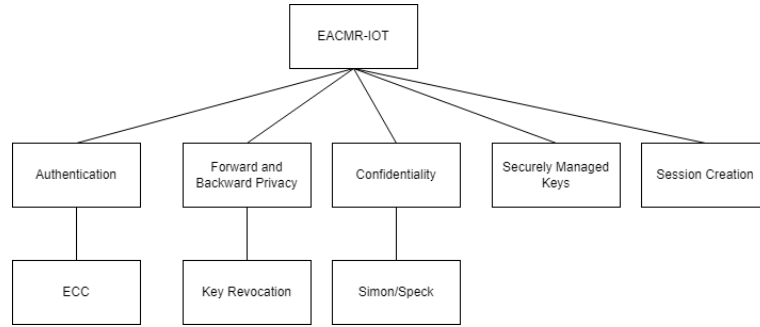


Figure 3.2: Components of EACMR-IoT

The implementation will be carried out on IoT networks operating with the RPL protocol, utilizing the **Contiki-NG operating system**. The **Cooja Simulator** will be used for testing and validation. The solution will aim to provide a secure communication protocol that not only meets the authentication and confidentiality needs of IoT networks but also guarantees forward and backward privacy, which is really important for maintaining security when nodes leave or join the network. This method will be scalable, resource-efficient, and compatible with existing IoT infrastructures, making it a practical solution for enhancing the security of RPL-based IoT networks.

Chapter 4

Conclusions and Future Work

EACMR-IoT addresses significant security weaknesses in the RPL-based IoT system by including strong cryptographic methods that are optimised for resource-limited settings. Furthermore, it employs the Elliptic Curve Cryptography (ECC) for robust authentication and the lightweight cryptography of Simon and Speck to guarantee its data secrecy. This integration not only improves security but also preserves resource efficiency, which is essential for IoT devices with restricted power and computing capacities.

In addition, the EACMR-IoT framework will provide a key revocation mechanism to effectively invalidate compromised keys, hence enhancing network security against potential data breaches. The implementation of the RPL protocol change in the Contiki-NG operating system demonstrates the effective utilisation of a secure routing protocol, thereby improving the security of IoT communications against unauthorised access and cyber threats in general. Furthermore, the system's forward and backward privacy measures guarantee that in the event of session keys being stolen, the security of both previous and future conversations remains uncompromised.

Conclusively, the EACMR-IoT framework offers a compact and effective solution that greatly enhances the security of IoT networks, making it highly suitable for implementation in important areas such as smart cities, healthcare, and military applications.

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